Zone Fall: Technical Sheet

Tech used

One self-contained HTML file (about 36 KB) with no external assets or libraries. HTML5 Canvas 2D rendering is DPR-aware (up to 2x) and resizes to any screen, portrait or landscape. All sound effects and music are generated in the browser with the Web Audio API (oscillators, filtered noise). Audio starts on the first tap, which iOS requires, and is skipped if the browser has no audio support. Stats are saved in localStorage. The simulation core is separate from the renderer, so it can run headless in Node for testing.

Game systems

- **World:** seeded procedural 3000x3000 island with up to 20 walled buildings with doors, about 170 trees and rocks, 4 lakes and around 120 loot spawns. The ground is pre-rendered once to an offscreen canvas.
- **Combat:** projectile bullets with sub-stepped collision, damage falloff past 70% of range, armor absorption, spread, magazine and reserve ammo, and reload time.
- **Zone:** 6 phases (wait, then shrink) with the next circle shown in advance and damage per second rising from 1 to 16.
- **Bot AI:** a state machine (loot, roam, fight, flee, heal, zone) that re-plans every 0.25-0.45 s. Bots have line-of-sight checks, per-bot skill (aim lead, turn rate, spread), strafing, loot scoring that compares weapon value and rarity, and auto-swap to their best gun.
- **Mobile UX:** floating twin sticks, big buttons sized to 15% of the short screen edge, aim assist, auto-reload, auto-pause when the app goes to the background, and a camera zoom adapted to the screen.
- **Effects:** muzzle flashes, additive-blend tracers and particles, screenshake, damage numbers, hit markers, damage-direction indicators, a pulsing glowing storm wall, swaying trees and see-through roofs.
- **Music:** a 16-step synth sequencer whose tempo rises from 96 to 146 BPM and adds layers as the zone shrinks.

Audit 1: Balance (simulated play)

I ran 30 full 30-bot matches headless after each tuning pass.

Pass	Avg length	Shortest	Winner weapon (R/SMG/SG/Sn/P)
v1 (20 matches)	3.8 min	0.9 min	11 / 7 / 0 / 1 / 1
v2	5.2 min	3.3 min	14 / 4 / 8 / 3 / 1
v3	4.7 min	2.5 min	6 / 5 / 10 / 8 / 1
Final	4.4 min	3.0 min	7 / 6 / 8 / 9 / 0

Fixes: shorter bot sight range (from 560-810 to 200-360) and target dropping, so fights no longer snowball in the first minute. A 40-second landing truce. Rifle nerfed (fire rate 5.5 to 4.1, damage 17 to 15). Shotgun buffed (damage 9 to 11, fire rate 1.1 to 1.25). SMG trimmed. Global damage scaled to 62%. Result: 4 viable weapons, with no single weapon taking more than 30% of wins. The storm causes about 10% of deaths, and matches run 3 to 8 minutes, which suits a phone session.

Audit 2: Bugs

The script passed node --check. A Node harness with a stubbed DOM and canvas played 6 full matches with a scripted thumb player (moving, looting, healing, swapping and firing at 60 fps) and logged 0 runtime errors. The player placed 5th to 22nd. Fixed: the minimap used destination-out compositing, which would have punched a hole through the whole game canvas, so it now uses an even-odd clip. Bot loot selection could loop on its own dropped gun, so loot is now only taken if it is at least 12% better. The door-gap collision let players pass through the top wall, so the gap is limited to the bottom wall. Inputs are cleared when the window loses focus, so a player no longer keeps running.

Audit 3: Player experience

Added: a landing-truce countdown, a "next circle" warning 15 seconds before the shrink, an off-zone banner with an arrow pointing back, damage-direction wedges, a GRAB button that lights up and names the item under it, an aim assist cone, auto-reload, auto-pause when the app is backgrounded, and first-time control hints.

Caveat: tested with simulated play, not on a physical phone.